Problem Set 10: Reactor Safety & Accident Analysis

Due: Friday, April 17, 2026

Instructions

- Show all work, including the equations used and unit conversions.
- **Constants & Data:**
  - **Decay Heat (Way-Wigner Formula):** The fractional power $P(t)/P_0$ after shutdown is approximated by:
    $$\frac{P(t)}{P_0} = 0.066\left[t^{-0.2} - (t+T_0)^{-0.2}\right]$$
    where t is time since shutdown (seconds) and T_0 is the reactor operating time (seconds).
  - **Reactivity Data:** Delayed Neutron Fraction $\beta = 0.0065$.
  - **Zirconium-Water Reaction:** $Zr + 2H_2O \rightarrow ZrO_2 + 2H_2 + Q$.
  - **Heat of Reaction:** Exothermic, $Q \approx 6.5$ MJ/kg of Zr.
  - **Hydrogen Combustion:** $2H_2 + O_2 \rightarrow 2H_2O$, $\Delta H_{comb} \approx 242$ kJ/mol H_2.

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1. **The Inevitable Heat (Decay Heat Physics)**
   A 3,000 MW_{th} reactor has been operating at full power for 18 months ($T_0 \approx 4.7 \times 10^7$ s) and scrams instantly at $t = 0$.
   (a) Using the Way-Wigner formula, calculate the thermal power produced by decay heat at:
   - $t = 1$ second (Immediate post-trip).
   - $t = 1$ hour (Short term cooling).
   - $t = 1$ month (Spent fuel pool timeframe).

   (b) At $t = 1$ hour, if the core is cooled by boiling water at atmospheric pressure ($h_{fg} \approx 2257$ kJ/kg), what is the boil-off rate in **kg/second**?
   (c) *Context:* A standard fire hose delivers ≈ 30 kg/s. Would a single fire truck be sufficient to keep this core covered one hour after shutdown?
   (d) **Critical Thinking (Reflood Danger):** In part (b), we assumed the cladding was relatively cool. If the core had been uncovered for some time and cladding temperatures exceeded 1200°C, would injecting water **immediately** cool the core? Explain why or why not, specifically referencing the chemical reaction detailed in Problem 4.

2. **Defense in Depth (Reliability Calculation)**
   Consider a Fukushima-type scenario. The probability of a "Station Blackout" (Loss of Offsite Power) is estimated at $f_{LOSP} = 0.1$ events/year. The plant has two redundant Emergency Diesel Generators (EDG-A and EDG-B) to provide backup power.

- Probability that EDG-A fails to start: $P(A) = 0.02$.
- Probability that EDG-B fails to start: $P(B) = 0.02$.

(a) Draw a simple Fault Tree for the "Total Loss of AC Power" top event. (Use logic gates: AND for independent failures, OR for redundant failures).

(b) Assuming the diesel failures are independent, calculate the annual frequency of a Total Loss of AC Power.

(c) **Common Mode Failure:** In the actual Fukushima event, the tsunami flooded the basement, disabling *both* generators simultaneously. If the probability of a tsunami exceeding the seawall is 10^{-3}/year, and it guarantees failure of both diesels ($P = 1.0$), what is the new total failure frequency? How does this compare to your answer in (b)?

3. The Positive Void Coefficient

The RBMK-1000 reactor had a positive void coefficient of reactivity, $\alpha_v \approx +2.0 \times 10^{-4} \frac{\Delta k/k}{\%\text{ void}}$.

(a) The reactor is running at a steady state. A pump failure causes the void fraction (steam bubbles) in the core to increase from 0% to 40%. Calculate the total reactivity insertion ρ (or $\Delta k/k$).

(b) Compare this reactivity insertion ρ to the delayed neutron fraction $\beta = 0.0065$.

(c) Based on the condition for Prompt Criticality ($\rho \geq \beta$), describe what happens to the reactor power period. Is the reactor controllable?

4. The Hydrogen Problem

During a severe accident, if the cladding temperature exceeds 1200°C, the Zirconium begins to react with steam. Assume a reactor core contains $20,000$ kg of Zirconium cladding. During the accident, 25% of the cladding oxidizes.

(a) Calculate the total mass of Hydrogen gas (H_2) produced (in kg).

(b) Calculate the chemical heat released by the Zirconium oxidation (in GJ). Compare this to the energy of 1 ton of TNT (4.18 GJ).

(c) If this hydrogen leaks into the reactor building and mixes with air, it can explode. Calculate the total energy released by the combustion of this hydrogen (in GJ).

(d) *Discussion:* Which energy source is larger: the heat from the metal oxidation (part b) or the hydrogen explosion (part c)?

5. Pressurizing the Containment

The ultimate safety barrier is the Containment Building. Consider a PWR Primary Coolant system containing $250,000$ kg of water at 300°C and 15 MPa. A Large Break LOCA occurs, releasing the entire coolant inventory into the Containment Building.

- Containment Volume $V = 50,000\ \text{m}^3$.
- Initial Containment Pressure = 1 atm (101 kPa).
- Initial Containment Temperature = 30°C.

(a) Assume the released water flashes to steam and reaches an equilibrium temperature of 120°C inside the containment. Using the Ideal Gas Law (assume steam behaves as an ideal gas at this low pressure), estimate the partial pressure of the steam.

(b) Calculate the total pressure in the containment (Air partial pressure + Steam partial pressure).

(c) If the design pressure of the containment is 4 atmospheres (absolute), does the building survive?